The Scout featured a lightweight airframe constructed primarily from composite materials, improving endurance and operational efficiency. It was powered by a small piston engine and equipped with a two-blade propeller, allowing flight durations of over 7 hours and a range exceeding 200 kilometers. Its stabilized electro-optical payload enabled high-resolution day and night imaging, while the telemetry and communication systems provided real-time data transmission to ground control stations. The Scout could take off and land from short, unprepared runways, offering remarkable deployment flexibility in field operations.

Operationally, the Scout proved its effectiveness during the 1982 Lebanon War, where it was used extensively for battlefield reconnaissance, artillery spotting, and electronic intelligence (ELINT). It provided live imagery of enemy movements, coordinated air and artillery strikes, and dramatically improved situational awareness for ground forces. This success demonstrated the tactical advantage of UAVs in coordinated military operations, inspiring numerous other countries to adopt similar systems (Zaloga, 2011).

Figure 3.6 depicts the IAI Scout (1979), one of the most influential UAVs of the second generation, developed by Israel Aerospace Industries (IAI). The figure highlights the Scout's twin-boom tail configuration, straight highmounted wings, and distinctive front-mounted propeller driven by a small piston engine. The compact and lightweight airframe, made of composite materials, enhanced endurance and operational efficiency, while the spherical electro-optical camera payload mounted beneath the nose enabled high-resolution real-time imaging. The illustration also shows the antenna housing atop the fuselage, representing its advanced data telemetry and communication system that allowed real-time video transmission to ground control stations. Operationally, the Scout was used for battlefield